MORE SETS, FUNCTIONS, AND RELATIONS

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1. Constructing New Sets from Old

Let A and B be sets.

Recall that we can construct new sets from A and B, namely

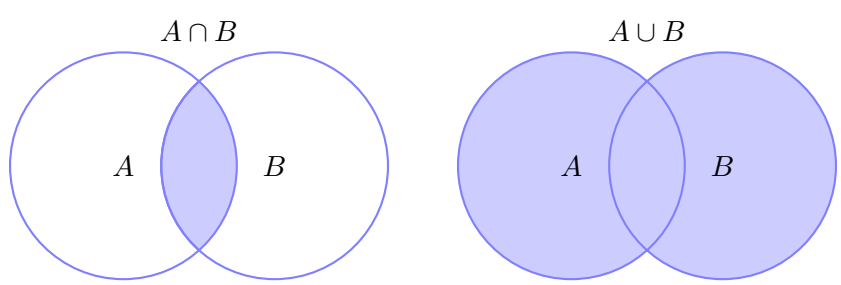
$$A \cup B = \{x : x \in A \text{ or } x \in B\} \qquad \text{“}A \textbf{ union } B\text{”}$$
$$A \cap B = \{x : x \in A \text{ and } x \in B\} \qquad \text{“}A \textbf{ intersect } B\text{”}$$
$$A \setminus B = \{x \in A : x \notin B\} \qquad \text{“}A \textbf{ complement } B\text{”}$$

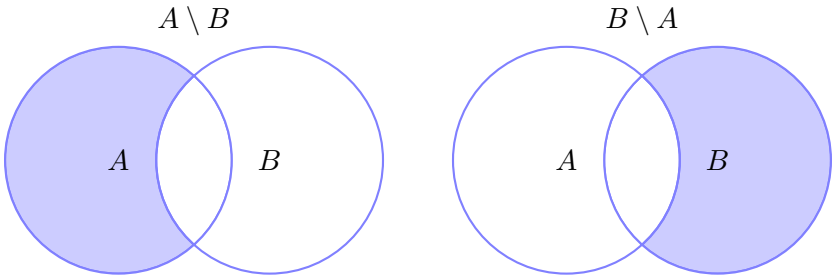
That is, the **union** is all objects that are contained in either set A or set B (or both!). Remember that the : and the | in sets are synonymous

The **intersection** is all objects that are in both A and B.

The **complement** is all objects that are in A but strictly not in B.

Pictorally, we have





One simple example of complement that will reappear regularly will be

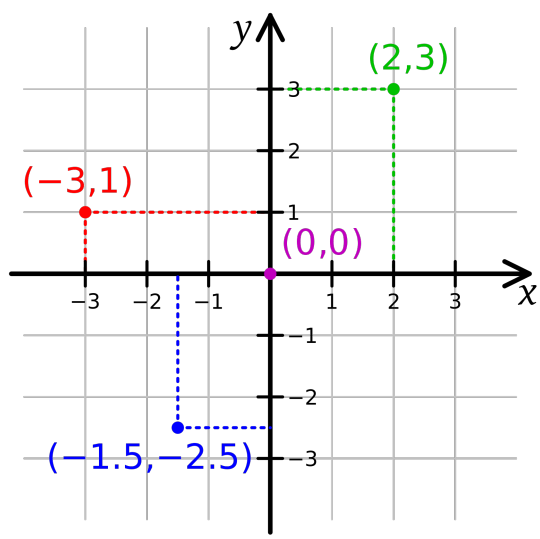
$$\mathbb{R} \setminus \{0\} = \{x \in \mathbb{R} : x \neq 0\} = \{\text{all non-zero real numbers}\}.$$

We define *the cartesian product* of A and B to be the set

$$A \times B \stackrel{\text{def}}{=} \big\{(a, b) : a \in A, b \in B\big\}.$$

The Cartesian product is the mathematical way of encoding **coordinates.**

Example 1. The xy plane is actually the cartesian product $\mathbb{R} \times \mathbb{R}$ (see picture below).



Recall that any point

Consider the cartesian product $\mathbb{R} \times \mathbb{R}$. This gives us the coordinate plane with the $x - y$ axis! It is called the $x - y$ plane since any point (x, y) in $\mathbb{R} \times \mathbb{R}$ can be described by identifying what x and y are.

Example 2. Let $A = \left\{ \text{☕}, \text{😄}, 2 \right\}$ and $B = \left\{ \emptyset, 0, \{1,2\}, 2 \right\}$. Then we get that

$$A \times B = \left\{ \begin{array}{cccc} (\text{☕}, \emptyset), & (\text{☕}, 0), & (\text{☕}, \{1,2\}), & (\text{☕}, 2) \\ (\text{😄}, \emptyset), & (\text{😄}, 0), & (\text{😄}, \{1,2\}), & (\text{😄}, 2), \\ (2, \emptyset), & (2, 0), & (2, \{1,2\}), & (2, 2) \end{array} \right\}.$$

$$A \cup B = \left\{ \text{☕}, \text{😄}, 2, \emptyset, 0, \{1,2\}, 2 \right\} = \left\{ \text{☕}, \text{😄}, 2, \emptyset, 0, \{1,2\} \right\}.$$

$$A \cap B = \{2\}$$

$$A \setminus B = \left\{ \text{☕}, \text{😄} \right\}$$

$$B \setminus A = \left\{ \emptyset, 0, \{1,2\} \right\}$$

Example 3. Recalling intervals, consider the cartesian product

$$[0,1] \times \{3\} = \left\{ (x,y) : 0 \leq x \leq 1, y = 3 \right\}$$

gives us a (horizontal) line segment from $(0,3)$ to $(1,3)$ (including the endpoints).

Example 4. (A quick exercise) Try to draw the following cartesian product in the (x,y) plane, i.e. $\mathbb{R} \times \mathbb{R}$

$$I = [0,1] \times (0,1).$$

Note that we can take as many Cartesian product as we want (finitely many, that is). For example, if we have $A_0, A_1, \ldots, A_n$ are all random sets, then we can still define their cartesian product (remember that the ordering in which you list the sets in the product matters!)

$$A_0 \times A_1 \times \cdots \times A_{n-1} \times A_n := \left\{ (x_0, x_1, \ldots, x_{n-1}, x_n) : x_i \in A_i \text{ where } 0 \leq i \leq n \right\}.$$

1.1. **Relations.** A *relation* between sets A and B is a subset $R \subseteq A \times B$ where

$$A \times B \stackrel{\text{def}}{=} \left\{ (a,b) : a \in A, b \in B \right\}$$

If $(a,b) \in R$, we write aRb instead.

If $A = B$ and R is a relation, i.e., a subset of $A \times A$, we then say "R is a relation on A."

What you're doing is just pick an object from A and then pick an object from B and put them in ( ) and saying these two objects are related (even if they come from different sets)

Example 5. We can define a relation, namely $\leq$, on $\mathbb{N}$, the natural numbers.

If $A, B = \mathbb{N}$ (from the definition above), then define R to be the set

$$R = \left\{(a, b) : a, b \in \mathbb{N}, \quad \text{and } a \leq b\right\}.$$

So in this case, we have $a \; R \; b$ if and only if $a \leq b$.

This example simply shows us that $\leq$ is a relation on $\mathbb{N}$.

Example 6. Let $A = B = \mathbb{R}$. Consider a relation $R := [0, 1] \times [5, 7)$. Then $a \in \mathbb{R}$ is related to $b \in \mathbb{R}$ if both conditions hold:

(1) $a \in [0, 1]$
(2) $b \in [5, 7)$.

Draw a visual in the cartesian plane of all of the $a, b \in \mathbb{R}$ such that aRb, i.e., all of the (x, y)-points inside the box with corners $(0, 5)$, $(0, 7), (1, 5)$ and $(1, 7)$ with an open border on $y = 7$ are the related points.

Example 7. Let $A, B = \mathbb{R}$ and let R be the set

$$R = \left\{(x, y) : x = y, \; x, y \in \mathbb{R}\right\}.$$

This is saying that xRy if and only if $x = y$. That is, we have just shown that equality is a relation on $\mathbb{R}$ (we will discuss about how equality is a special type of relation, called *equivalence relation*).

2. Functions

What are functions? It is a way of relating two sets (not necessarily distinct) to each other! Think of it as a machine (e.g. your computer) that takes in user inputs (pressing the SHIFT key and the "`q`" key) and spits out a single well-defined output (a upper case `Q`).

Let's put a formal definition to this: Let A and B be sets. A *function* f with *domain* A and *codomain* B, denoted $f : A \to B$ is a relation $R_f \subseteq A \times B$ such that the two extra conditions hold:

(1) Defined on domain: For all $a \in A$, there exists a $b \in B$ such that $(a, b) \in R_f$ (i.e., every domain point/input has an output value)
(2) Vertical line test: If (a, b) and $(a, b') \in R_f$, then $b = b'$ (i.e., there is exactly one output for a given input)

If $(a, b) \in R_f$, then we write $f(a) = b$.

Let's look at some examples of the *failures* of condition (1) and (2), respectively.

For failing (1), consider

$$f(x) = \frac{1}{x}.$$

We know that we cannot say the domain of f is $\mathbb{R}$ since $0 \in \mathbb{R}$ and there is no output value for $x = 0$. Thus, saying "$f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = 1/x$" is incorrect. Instead, we should say "$f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be a function defined by $f(x) = 1/x$", since the domain is all real numbers except zero.

For the failure of (2), we can look at vertical lines of the graph of f to determine whether or not f is a function. If you are given a table of (x, y) values however (that table is a relation!), (2) says that if there is any repetitions in the x place found in the table, the y values should match otherwise that relation is *not* a function.

Example 1. Let $S = \{1, 2, 3\}$ and $T = \{0, 1\}$. Define a function $f : S \to T$ by letting $f(1) = 0, f(2) = f(3) = 1$.

(1) What is the domain of f?
(2) What is the codomain of f?
(3) What is the relation R_f defining f? (Write a set of (x, y)-values)

(See footnote[1] for solution.)

The *range* (or *image*) of a function $f : A \to B$ is the subset of B

$$\mathrm{Im}(f) \stackrel{\text{def}}{=} \left\{ f(a) \in B : a \in A \right\}.$$

Example 2.

(a) Let $A = \{x, y, z\}$ and $B = \{1, 2, 3\}$. Define a function $f : A \to B$ by $f(x) = f(y) = 1$, and $f(z) = 2$:

$$\begin{tikzcd}
A \arrow[r] & B \\
x \arrow[r] & 1 \\
y \arrow[ur] & 2 \\
z \arrow[ur] & 3
\end{tikzcd}$$

and so we have

$$\begin{aligned} \text{domain of } f: &\quad A = \{x, y, z\} \\ \text{codomain of } f: &\quad B = \{1, 2, 3\} \end{aligned}$$

[1]Domain is $S = \{1, 2, 3\}$, the codomain is $T = \{0, 1\}$ and the relation is $R_f = \left\{(1, 0), (2, 1), (3, 1)\right\}$

$$\text{image/range of } f: \quad \{1, 2\}$$

(b) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 3$. Then we have

$$\begin{aligned} \text{domain of } f: &\quad \mathbb{R} \\ \text{codomain of } f: &\quad \mathbb{R} \\ \text{image/range of } f: &\quad \{3\} \end{aligned}$$

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x^2$. Then we have

$$\begin{aligned} \text{domain of } f: &\quad \mathbb{R} \\ \text{codomain of } f: &\quad \mathbb{R} \\ \text{image/range of } f: &\quad [0, \infty) \end{aligned}$$

(d) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 9x$. Then we have

$$\begin{aligned} \text{domain of } f: &\quad \mathbb{R} \\ \text{codomain of } f: &\quad \mathbb{R} \\ \text{image/range of } f: &\quad \mathbb{R} \end{aligned}$$

(e) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be defined by $f(x) = 9/x$. Then we have

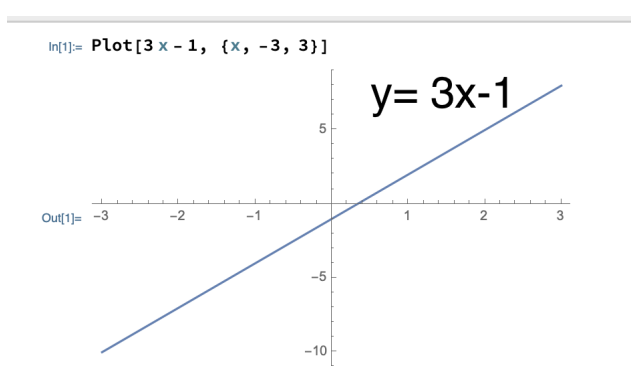
$$\begin{aligned} \text{domain of } f: &\quad \mathbb{R} \setminus \{0\} \\ \text{codomain of } f: &\quad \mathbb{R} \\ \text{image/range of } f: &\quad \mathbb{R} \end{aligned}$$

2.1. **For Fun (graphs as sets).** We know for a fact that functions, specifically lines, are defined by their graphs (for lines, they are characterised by their slope and y-intercept). We can give the graph of a function a rigourous mathematical definition as well:

Let $f : A \to B$ be a function (take $A = B = \mathbb{R}$ for ease of pictures). Then the *graph of* f is the set

$$\Gamma(f) \stackrel{\text{def}}{=} \{(x, y) : y = f(x)\}.$$

Example 3. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = 3x - 1$.



The blue line, which is exactly what $f(x)$ is, is technically $\Gamma(f)$ where $f(x) = 3x - 1$. That is,

$$\Gamma(f) = \left\{(x, 3x - 1) : x \in \mathbb{R}\right\}.$$